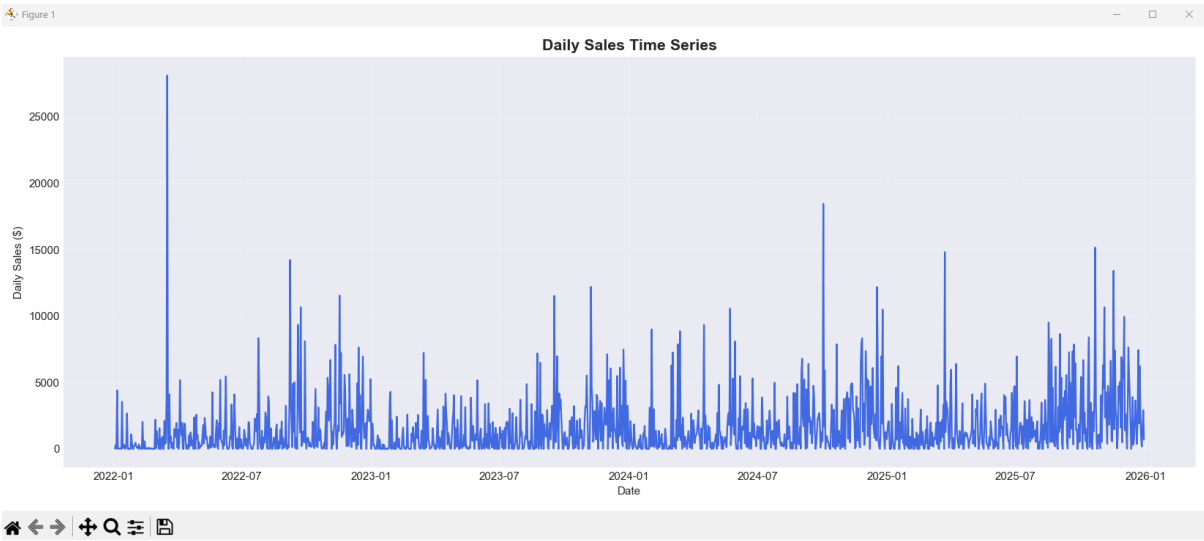
ARIMA Model Output

1)

```
============================================================
TIME SERIES PREPARATION
============================================================

☑ Daily sales aggregated
Total days in dataset: 1458
Date range: 2022-01-03 00:00:00 to 2025-12-30 00:00:00
Total sales: $2,261,536.78
Average daily sales: $1,551.12
Median daily sales: $797.93
Std daily sales: $2,211.50
```



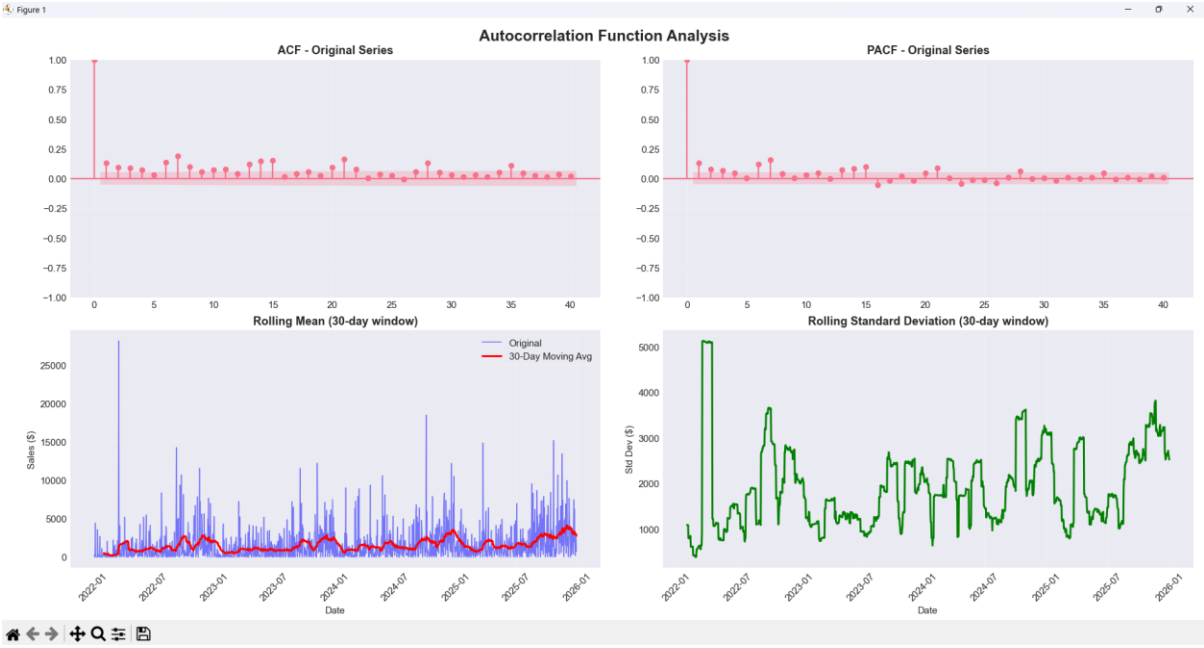
2)

```
======================================================================
STATIONARITY ANALYSIS
======================================================================

📊 Analyzing: Original Sales Series
ADF Statistic: -4.9254
p-value: 0.000031
☑ Original Sales Series is STATIONARY (reject null hypothesis)

☑ No differencing needed (d=0)

📌 Final differencing order: d=0
```



3)

```
======================================================================
ARIMA MODEL TRAINING
======================================================================

📊 Data Split:
Training: 2022-01-03 00:00:00 to 2025-03-13 00:00:00 (1166 days)
Testing: 2025-03-14 00:00:00 to 2025-12-30 00:00:00 (292 days)
Train/Test ratio: 80.0%/20.0%

🔍 Testing common ARIMA models (d=0)...
  ARIMA(1, 0, 1): AIC=21108.12
  ARIMA(0, 0, 1): AIC=21138.84
  ARIMA(1, 0, 0): AIC=21137.23
  ARIMA(2, 0, 2): AIC=21109.84
  ARIMA(1, 0, 2): AIC=21108.97
  ARIMA(2, 0, 1): AIC=21108.94
  ARIMA(0, 0, 0): AIC=21152.17
  ARIMA(3, 0, 3): AIC=21089.30

☑ Best ARIMA order: ARIMA(3, 0, 3)
☑ Best AIC: 21089.30

🏋 Training ARIMA model...
☑ Model trained successfully!
```

```
📋 Model Summary:
------------------------------------------------------------
                               SARIMAX Results
==============================================================================
Dep. Variable:                  Sales   No. Observations:                 1166
Model:                 ARIMA(3, 0, 3)   Log Likelihood              -10536.652
Date:                Mon, 20 Apr 2026   AIC                          21089.304
Time:                        12:29:49   BIC                          21129.795
Sample:                    01-03-2022   HQIC                         21104.578
                         - 03-13-2025
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const       1388.2256    271.908      5.105      0.000     855.296    1921.155
ar.L1          0.5300      0.018     28.670      0.000       0.494       0.566
ar.L2         -0.5543      0.016    -33.802      0.000      -0.586      -0.522
ar.L3          0.9569      0.021     46.130      0.000       0.916       0.998
ma.L1         -0.4804      0.031    -15.489      0.000      -0.541      -0.420
ma.L2          0.5603      0.030     18.902      0.000       0.502       0.618
ma.L3         -0.8804      0.033    -26.592      0.000      -0.945      -0.816
sigma2      4.239e+06    6.8e+04     62.310      0.000    4.11e+06    4.37e+06
===================================================================================
Ljung-Box (L1) (Q):                   2.22   Jarque-Bera (JB):             63864.01
Prob(Q):                              0.14   Prob(JB):                         0.00
Heteroskedasticity (H):               0.90   Skew:                             4.23
Prob(H) (two-sided):                  0.30   Kurtosis:                        38.25
===================================================================================
```

```
============================================================
MODEL EVALUATION
============================================================


------------------------------------------------------------

📊 Training Set Metrics:
   Mean Absolute Error (MAE): $1,284.98
   Mean Squared Error (MSE): $4,133,885.45
   Root Mean Squared Error (RMSE): $2,033.20
   R² Score: 0.0627
   Mean Absolute Percentage Error (MAPE): 766.55%
   Forecast Accuracy: 0.00%

------------------------------------------------------------

📊 Test Set Metrics:
   Mean Absolute Error (MAE): $1,687.09
   Mean Squared Error (MSE): $6,875,714.13
   Root Mean Squared Error (RMSE): $2,622.16
   R² Score: -0.0977
   Mean Absolute Percentage Error (MAPE): 453.52%
   Forecast Accuracy: 0.00%
```

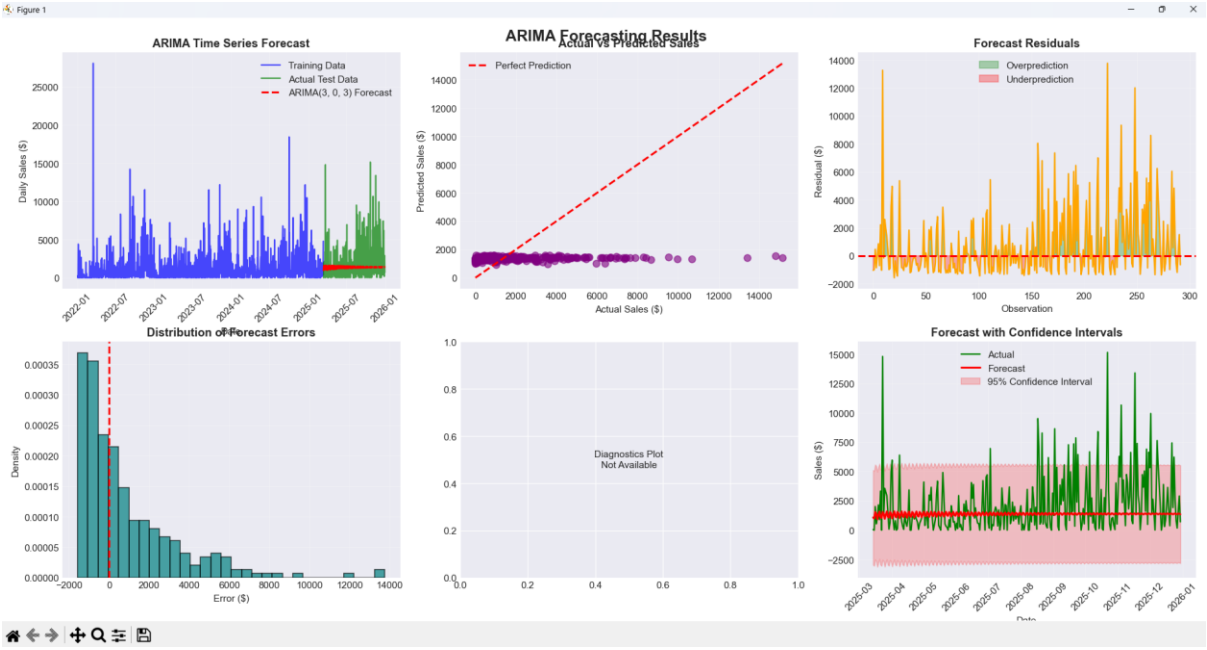


Figure 1
ARIMA Forecasting Results
ARIMA Time Series Forecast
Training Data
Actual Test Data
ARIMA(3, 0, 3) Forecast
Daily Sales ($)
Actual vs Predicted Sales
Perfect Prediction
Predicted Sales ($)
Actual Sales ($)
Forecast Residuals
Overprediction
Underprediction
Residual ($)
Observation
Distribution of Forecast Errors
Density
Error ($)
Diagnostics Plot
Not Available
Forecast with Confidence Intervals
Actual
Forecast
95% Confidence Interval
Sales ($)

4)

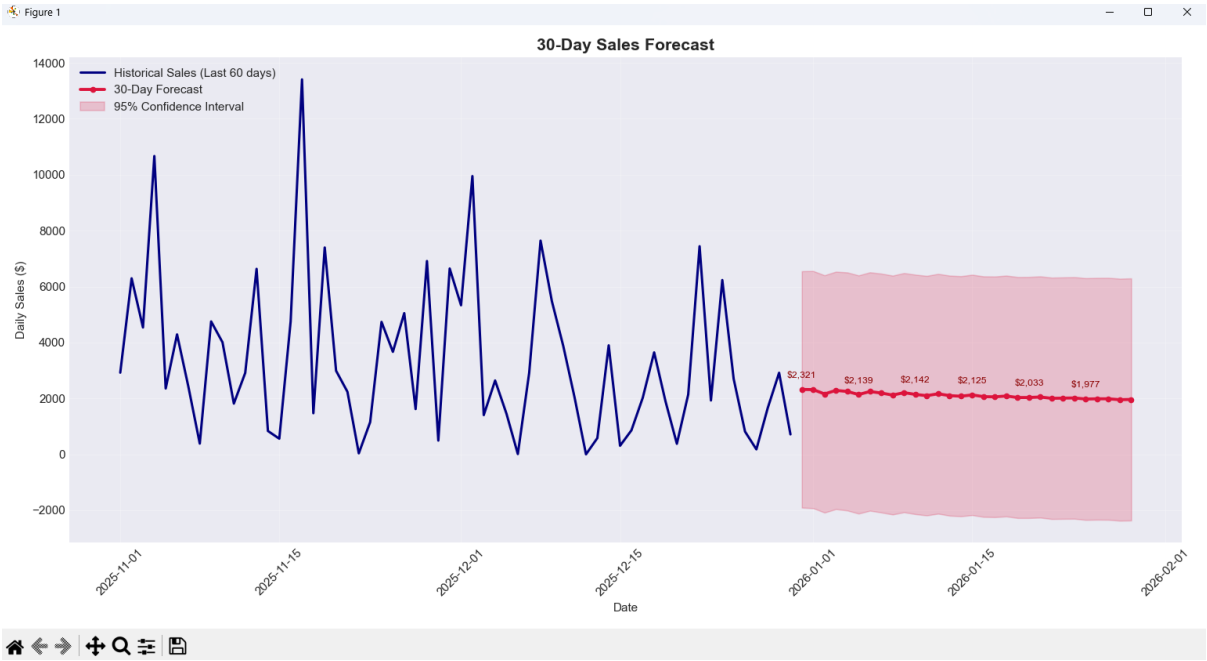


Figure 1
30-Day Sales Forecast
Historical Sales (Last 60 days)
30-Day Forecast
95% Confidence Interval
Daily Sales ($)
$2,321
$2,139
$2,142
$2,125
$2,033
$1,977
Date

5)

📅 30-DAY SALES FORECAST REPORT
==========

📊 First 10 Days Forecast:

Date	Forecasted_Sales	Lower_95%_CI	Upper_95%_CI
2025-12-31	$2,320.97	$-1,907.13	$6,549.08
2026-01-01	$2,314.73	$-1,927.07	$6,556.52
2026-01-02	$2,159.73	$-2,083.79	$6,403.25
2026-01-03	$2,285.67	$-1,961.56	$6,532.89
2026-01-04	$2,248.21	$-2,009.35	$6,505.76
2026-01-05	$2,139.14	$-2,119.89	$6,398.17
2026-01-06	$2,246.56	$-2,016.69	$6,509.82
2026-01-07	$2,191.06	$-2,079.93	$6,462.05
2026-01-08	$2,119.07	$-2,153.31	$6,391.45
2026-01-09	$2,205.70	$-2,071.11	$6,482.50

💡 KEY INSIGHTS:
1. Average daily forecast: $2,105.05
2. Total forecasted sales: $63,151.45
3. Peak forecast: $2,320.97
4. Minimum forecast: $1,953.51

💾 Forecast saved to 'sales_forecast_arima.csv'

==========
✅ FORECASTING COMPLETE!
==========

📊 FINAL SUMMARY:
Dataset size: 1458 days
Best ARIMA model: ARIMA(3, 0, 3)
Test set size: 292 days
Test R^2 Score: -0.0977